

Where we are

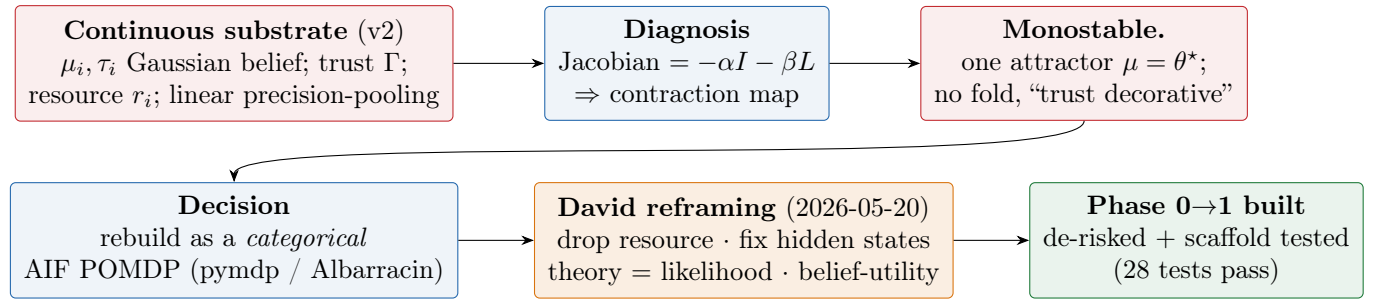
A visual recap of the paradigm-dynamics rebuild
(continuous substrate $\rightarrow$ categorical epistemic POMDP)

Internal recap — IWAI 2026

2026-05-21

One-sentence state: we abandoned the continuous Gaussian model (provably monostable), decided to rebuild as a categorical active-inference POMDP, simplified it after a conversation with David (drop resource, fix the hidden states, make “theory” a likelihood, make *belief-utility* the core), de-risked the bistability question, and built+tested the Phase-1 scaffold — which surfaced one important wall (next page).

1. The arc: what changed and why

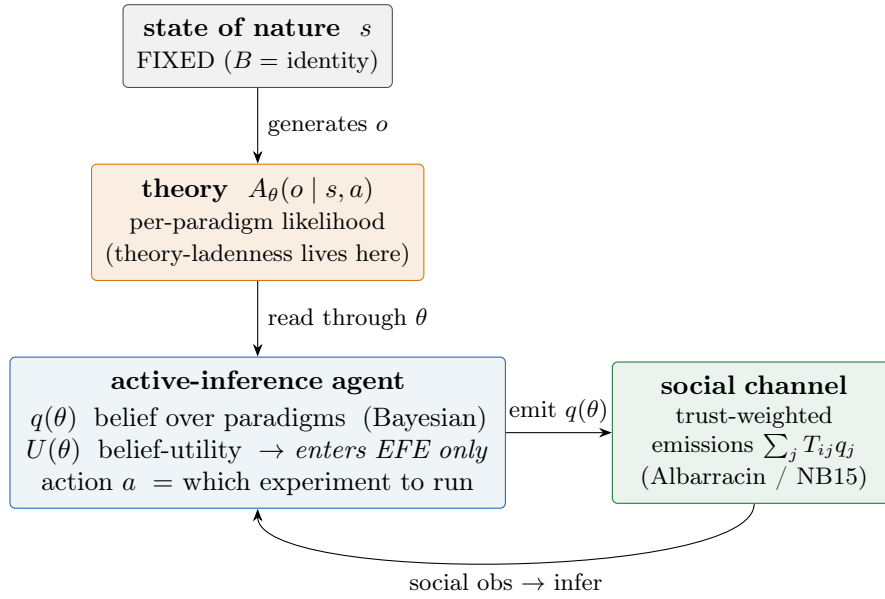


The honest justification for going categorical is *not* that it was forced (the monostability proof is for the linearised reduction, and the literature lists six bistability mechanisms). It is **strategic**: literature alignment with Albarracin’s “Epistemic Communities”, and the POMDP exposes the order parameters David asked for natively ($q(\theta)$, expected free energy, $q(\pi)$).

2. Old model vs. new model (what each object became)

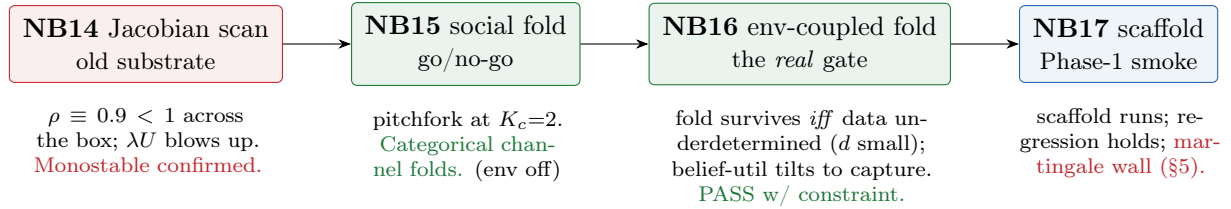
	Old (continuous v2)	New (categorical POMDP)
belief over theory	Gaussian μ_i, τ_i over $\theta \in \mathbb{R}$	categorical $q_i(\theta)$, $\theta \in \{A, B\}$
how data is read	one shared regression likelihood	per-paradigm likelihood $A_\theta(o s, a)$ (<i>theory-ladenness</i>)
truth	moving target $\theta^*(t)$	fixed state of nature; $B = \text{identity}$
actions	choose experiment x (info-gain)	choose experiment a (<i>measurement, not control</i>)
social coupling	linear precision-pool (DeGroot)	softmax-precision channel (<i>folds, NB15</i>)
“mind-change” / motivation	$\lambda \cdot U$ in the <i>update</i> (blew up)	belief-utility $U(\theta)$ in the <i>EFE/policy</i>
resource	r_i flow + inflow (a state)	dropped (was a slaved read-out)

3. The new generative model (one agent)

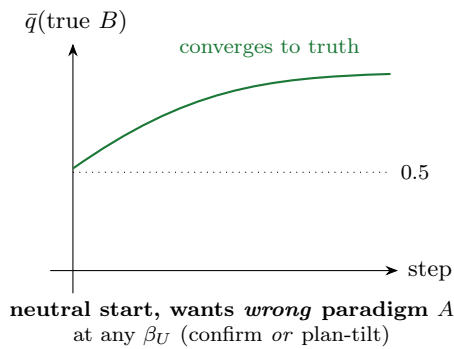


Two engines were meant to drive lock-in: (i) *motivated sampling* — $U(\theta)$ biases which experiments you run; (ii) *social reinforcement* — the NB15 softmax-precision fold. Belief-utility heterogeneity was meant to be the symmetry-breaking field turning polarization into *capture*. Section 5 is where that ran into a wall.

4. What the four notebooks established



5. The Phase-1 finding: the martingale wall



Belief-utility in the policy cannot flip the basin.

By the martingale property $\mathbb{E}_o[\text{posterior}] = \text{prior}$, honest Bayesian inference against a truthful world converges to the truth no matter what the agent *wants*.

Capture in the scaffold comes from the **social fold + initial conditions**, *not* from belief-utility. (Same wall the NB14 λU blow-up signalled.)

6. The open decision (for you / David)

Where should genuine *capture-against-evidence* come from? The scaffold supports all three with localised changes.

1. Theory-ladenness

per-agent A_θ : commit to a paradigm $\Rightarrow$ interpret data through *its* likelihood $\Rightarrow$ honest Bayes captures.
most faithful to option- B ; A_θ currently shared

2. Bounded belief-field

NB16's h_U on the belief (touches the update). Validated in NB16; needs bounding vs the NB14 blow-up.

3. Social-fold only

belief-utility stays exploration-only; capture is sociological — who you start near, who you trust (asymmetric IC / trust).

Built this round (all in `src/pomdp/`, 28 tests pass): `gen_model.py` (per-paradigm A_θ , $B=\text{id}$, $C/D/U$), `agent_pop.py` (categorical AIF + belief-utility EFE term, 3 modes), `step.py` (emit $\rightarrow$ route $\rightarrow$ infer $\rightarrow$ act), `observables.py` (occupancy, EFE/ $q(\pi)$, polarization, capture). Notebooks 14–17. Resource design archived in `notes/archive/`. Full spec: `notes/pomdp_generative_model.md`.